Vraj Patel

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EDUCATION

Queen's University Kingston, ON
BASc, Computer Engineering *Sep 2026 - Apr 2030*

PROJECTS

LiDAR SLAM Mecanum Rover | *ROS 2, micro-ROS* Jul - Aug 2026

- Generated and saved an occupancy-grid map with RPLIDAR C1 and slam_toolbox, tracking the rover's position and orientation while driving.
- Used an RP2040 for real-time rover motor control and a Raspberry Pi 5 for SLAM, linked through serial micro-ROS.
- Programmed mecanum wheel control for forward motion, strafing, and rotation, with four wheel encoders and an IMU feeding rover odometry.

ESP32 Quadcopter | *C++, ESP32* Jul - Aug 2026

- Built a custom quadcopter controller around an F310 gamepad and ESP32-S3, sending throttle, roll, pitch, yaw, and arm/disarm commands over ESP-NOW.
- Assembled an F450 quadcopter running esp-fc firmware on an ESP32 for stabilized takeoff, hover, and manual flight.
- Wired an MPU-6500 IMU, BMP280 barometer, and QMC5883L magnetometer to the quadcopter's ESP32 over I2C for motion, pressure, and heading measurements.

6-DOF Robotic Arm | *Python, Arduino* Jan - Mar 2026

- Wrote a Python inverse-kinematics solver for a 6-DOF arm, searching gripper pitch to resolve redundancy while rejecting joint-limit violations and table collisions.
- Calibrated robotic-arm servos using two-pose linear fits to convert inverse-kinematics joint angles into servo commands, interpolated over serial for smooth motion.
- Designed and assembled a 3D-printed arm with six servos, including a mirrored shoulder pair, and a NEMA-17 stepper base.

Guitar Effects Pedal PCBs | *KiCad* Sep - Nov 2025

- Created KiCad layouts and Gerber files for Boss DS-1, Big Muff Pi, and EHX LPB-1 pedals from schematics.
- Soldered and debugged three working through-hole pedals, applying gain staging, clipping, and biasing.

Other Projects: Gesture-Controlled Robotic Hand (Python, Arduino); Hybrid Quadruped Robot (Arduino Nano ESP32).

EXPERIENCE

Founder & Captain, FTC Teams #27964 & #32514 Sep 2023 - Jun 2026
William Lyon Mackenzie CI *Toronto, ON*

- Founded a school robotics program with a general robotics club and appointed technical leads, growing membership to 50 across two FTC teams.
- Led the FTC mechanical sub-team, designing the mecanum drivetrain and manipulator arm in Fusion 360 and assembling the school's first competition robot in two months.
- Led weekly Arduino workshops and school outreach, introducing 100+ students to electronics and programming.

Database Designer (Co-op) Sep 2024 - Feb 2025
Access Group Inc. *Toronto, ON*

- Automated Intune, Defender, and remote-monitoring reports with PowerShell and Power BI, replacing manual Excel reports.
- Built Power BI dashboards from ConnectWise API data, giving IT staff access to account information for 100 users.

AWARDS

1st Place in Canada - CyberTitan 2026
National cybersecurity competition; six-person team; Windows system hardening.

TECHNICAL SKILLS

Design & Simulation: KiCad, Fusion 360, SolidWorks, Onshape, LTspice
Embedded Hardware: ESP32, RP2040, STM32, Raspberry Pi
Fabrication & Prototyping: Soldering, 3D Printing
Programming: C, C++, Python, PowerShell
Robotics & Controls: ROS 2, micro-ROS, slam_toolbox, PID Control, Robot Kinematics
Software Tools: Linux, Git/GitHub, MATLAB